Dr. Jinxin Li

Physicist & Software Engineer

Scientific Computing · C++ / Python · AI Systems

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Professional Profile

Experimental particle physicist and software engineer with nearly seven years of experience developing and maintaining C++ and Python software for scientific computing, simulation and data analysis. Combines quantitative problem-solving with hands-on algorithm development, complex framework integration, debugging and automated validation in Linux/HPC environments. Applied this expertise to AI systems by architecting PANDA Agent, an independent research assistant integrating code-aware retrieval, backend workflows, and systematic evaluation.

Technical Skills

Programming & Engineering: C++, Python, Bash/Shell, object-oriented design, Git, Linux, Docker, debugging, software integration, automated testing.

AI & Backend Systems: RAG, LangGraph, FastAPI, REST APIs, PostgreSQL, Qdrant, SQLAlchemy, hybrid dense/BM25 retrieval, reranking, evaluation.

Scientific Computing & HPC: ROOT, PandaRoot, FairRoot, NumPy, Pandas, Scikit-learn, SLURM, Singularity, Monte Carlo simulation, statistical analysis.

Languages: Chinese (Native), English (Proficient), German (CEFR A2).

Professional Experience

Doctoral Researcher / Scientific Software Developer

Ruhr University Bochum - Institute of Experimental Physics I, PANDA Collaboration
Sep. 2019 – Jul. 2026 · Bochum, Germany

- Developed and maintained C++ and Python software for detector simulation, particle reconstruction, numerical modeling and scientific data analysis within the PandaRoot ecosystem.
- Served as Software Owner for the PANDA Luminosity Detector Group and primary maintainer of the existing LuminosityFit C++ framework, responsible for maintenance, debugging, functional extensions and integration.
- Implemented and integrated reconstruction algorithms and simulation-based correction workflows, contributing cross-component changes and bug fixes to PandaRoot.
- Developed repeatable validation workflows and technical documentation, and supported collaborative research through software training and undergraduate thesis mentoring.

Selected Engineering Projects

PANDA Agent - AI Research Assistant & Code Intelligence

Apr. 2026 – Present · Independent project · github.com/JinxinLee/PANDA_Agent

- Architected a local AI research assistant integrating scientific source code, literature and documentation through version-aware ingestion, hybrid dense/BM25 retrieval, reranking and evidence-grounded question answering.
- Designed and integrated Python backend workflows using LangGraph, FastAPI, PostgreSQL, and Qdrant, with structured diagnostics, reproducible execution workflows, and systematic validation.
- Established versioned evaluation and regression testing using human-reviewed benchmarks, controlled ablations and systematic failure analysis to assess retrieval and answer quality.

Selected Engineering Projects (continued)

LuminosityFit - C++ Framework Development

Mar. 2022 – Jul. 2024 · Doctoral research

- Extended an existing C++ fitting framework to model distributed interaction regions and beam parameters, debugging and validating new functionality integrated with PandaRoot.
- Integrated reconstructed target distributions into an end-to-end reconstruction and fitting chain, achieving **(0.05 ± 0.07)% relative luminosity deviation** from simulated ground truth under the studied conditions.

Particle Reconstruction & Iterative Correction

Jul. 2024 – Apr. 2026 · Doctoral research

- Developed a POCA-based vertex reconstruction workflow, integrating existing tracking and GenFit/Kalman components within PandaRoot.
- Implemented simulation-based efficiency corrections and iterative refinement, reducing global target-profile discrepancy from **12.27% after iteration 1 to 1.30% after iteration 3** in a controlled simulation study.

Detector Simulation & Performance Evaluation

Sep. 2019 – Mar. 2022 · Doctoral research

- Implemented an alternative HV-MAPS detector geometry in PandaRoot and benchmarked reconstruction performance against the baseline detector configuration.
- Benchmarked alternative detector configurations using Monte Carlo simulation, with the HV-MAPS design achieving **2.8 times the simulated D-meson reconstruction efficiency of the baseline (9.26% vs. 3.33%)**.

Education

Dr. rer. nat. (Ph.D.) in Physics

Ruhr University Bochum, Germany · 2026

Research: Luminosity determination, particle reconstruction and detector simulation for the PANDA experiment.

B.Sc. in Physics

Nankai University, Tianjin, China · Sep. 2015 – Jun. 2019

Earlier Research Experience

IHEP, Chinese Academy of Sciences / BESIII

Research Assistant · 2018 – 2019 · Charm/D-meson analysis and Monte Carlo simulation.

Nankai University / BESIII

Undergraduate Research Assistant · 2016 – 2019 · Charmonium analysis and scientific data processing.

AMS-02 Project (CERN / MIT collaboration)

Summer Research Assistant · 2017 · Detector simulation and tracker-resolution validation.